

Sequence Comparison Tools

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1 Scope

This folder contains nine notebooks that start from the same FASTA inputs and represent sequence relationships in different mathematical ways:

- ternary_alignment_substitution.ipynb
- kras_msa_publication.ipynb
- sequence_similarity_network.ipynb
- proteinmpnn_homologous_pdb.ipynb
- ternary_hamming.ipynb
- ternary_levenshtein.ipynb
- trilateration_hamming.ipynb

- `mds_levenshtein.ipynb`
- `kmer_pca.ipynb`

The first notebook is a reference-based ternary map built from alignment scores, the second is a publication-oriented MSA visualization workflow, the third is a sequence similarity network analysis, the fourth is a structure-conditioned sequence design workflow with ProteinMPNN, the fifth and sixth are simpler reference-based ternary maps, the seventh is a reference-based trilateration embedding, the eighth is a global distance-matrix ordination, and the ninth is a feature-based compositional ordination.

2 Input model

All notebooks use the same logical setup:

- a FASTA file with exactly three reference sequences A , B , and C ,
- a FASTA file with additional query sequences $Q_1, \dots, Q_n$.

For any query Q , the basic reference-based quantities are

$$d_A(Q) = d(Q, A), \quad d_B(Q) = d(Q, B), \quad d_C(Q) = d(Q, C),$$

or, in similarity form,

$$s_A(Q), \quad s_B(Q), \quad s_C(Q).$$

Depending on the notebook, these quantities are turned into:

- a ternary composition of relative similarity,
- a trilaterated point in the plane,
- a full pairwise distance matrix for ordination,
- or a feature vector built from k -mer compositions.

3 Distance and similarity functions

This section summarizes the main sequence-comparison models used across the notebooks, following standard bioinformatics treatments of sequence distance, alignment, and scoring [15].

3.1 Normalized Hamming distance

If two sequences X and Y have the same length L , the normalized Hamming distance is

$$d_{\text{Ham}}(X, Y) = \frac{1}{L} \sum_{k=1}^L \mathbf{1}\{X_k \neq Y_k\}.$$

This is the fraction of mismatching positions. It is simple and interpretable, but requires aligned or at least position-wise comparable sequences of equal length, and it gives every mismatch the same cost.

3.2 Normalized Levenshtein distance

For sequences of arbitrary lengths, let $\text{Lev}(X, Y)$ denote the edit distance, i.e. the minimum number of insertions, deletions, and substitutions needed to transform X into Y . The normalized form used in the notebooks is

$$d_{\text{Lev}}(X, Y) = \frac{\text{Lev}(X, Y)}{\max(|X|, |Y|)}.$$

This allows indels and makes distances comparable across different lengths, but substitutions still all have unit cost.

3.3 Normalized alignment distance with a substitution matrix

For homologous sequences, especially proteins, a more informative comparison is obtained from an optimal pairwise alignment score $S(X, Y)$ computed with

- a global alignment algorithm such as Needleman–Wunsch when full-length homology is expected [16],
- or a local alignment algorithm such as Smith–Waterman when only a conserved region should drive the comparison [19],
- a substitution matrix,
- affine gap penalties.

For proteins, the standard practical default is BLOSUM62 with affine gaps [12]. If one wants an explicit divergence-controlled family of matrices instead, the classical PAM construction is the standard alternative [7]. A strong normalized dissimilarity is

$$d_{\text{align}}(X, Y) = 1 - \frac{S(X, Y)}{\sqrt{S(X, X) S(Y, Y)}}.$$

Equivalently,

$$s_{\text{align}}(X, Y) = \frac{S(X, Y)}{\sqrt{S(X, X) S(Y, Y)}}$$

is a normalized alignment similarity. This normalization ensures $d_{\text{align}}(X, X) = 0$ and partially adjusts for sequence length and composition through self-scores.

An alternative baseline normalization is

$$d(X, Y) = 1 - \frac{S(X, Y) - S_{\min}}{S_{\max} - S_{\min}},$$

but the self-score normalization above is often the simplest useful default in practice.

3.4 Why alignment-based similarity is often better for proteins

Hamming and Levenshtein treat all substitutions equally. Biologically, however, an Ile↔Leu replacement is usually milder than a Trp↔Gly replacement. A substitution matrix incorporates these differential costs. Therefore alignment-based similarity is typically preferable for proteins whenever the sequences are alignable, and structure-based benchmarking has long shown the importance of comparison schemes that respect distant evolutionary relationships [4].

3.5 Sequence-type recommendations

- For proteins: global alignment with BLOSUM62 and affine gaps is the default recommendation for the ternary workflow.
- For DNA/RNA: a substitution matrix can be used operationally, but model-based nucleotide distances such as Jukes–Cantor or Kimura are often more principled when the goal is evolutionary interpretation.
- For highly diverged, rearranged, recombinant, or very large sequence collections: feature-based or alignment-free methods may be more robust.

3.6 Important caveat

A raw alignment score is not automatically a metric. Triangle inequality may fail, normalization choices matter, and gap penalties affect geometry. Moreover, the statistical interpretation of alignment scores in sequence search is a separate issue from geometric embedding, as emphasized in scoring-significance theory and database-search methods [13, 2]. For the visual tools in this folder, what is required is a consistent dissimilarity, not necessarily a strict metric in the formal sense.

4 Notebook 1: ternary mapping from normalized alignment distance

4.1 Distance-to-similarity model

The notebook `ternary_alignment_substitution.ipynb` computes pairwise alignment scores with Biopython’s `PairwiseAligner`, affine gap penalties, and the substitution matrix BLOSUM62. The notebook is now intentionally specialized to homologous protein sequences and uses a bundled K-RAS protein example by default.

For any query Q , define

$$s_A(Q) = s_{\text{align}}(Q, A), \quad s_B(Q) = s_{\text{align}}(Q, B), \quad s_C(Q) = s_{\text{align}}(Q, C),$$

where

$$s_{\text{align}}(X, Y) = \frac{S(X, Y)}{\sqrt{S(X, X) S(Y, Y)}}.$$

The corresponding distance is

$$d_i(Q) = 1 - s_i(Q), \quad i \in \{A, B, C\}.$$

4.2 Closure into a ternary composition

The three normalized similarities are closed into

$$\lambda_i(Q) = \frac{s_i(Q)}{s_A(Q) + s_B(Q) + s_C(Q)}, \quad i \in \{A, B, C\}.$$

Hence

$$\lambda(Q) = (\lambda_A(Q), \lambda_B(Q), \lambda_C(Q))$$

is a 3-part composition satisfying

$$\lambda_A(Q) + \lambda_B(Q) + \lambda_C(Q) = 1, \quad \lambda_i(Q) \geq 0.$$

4.3 Barycentric map into the triangle

The triangle vertices are fixed as

$$V_A = (0, 0), \quad V_B = (1, 0), \quad V_C = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

The plotted point is

$$\mathbf{p}(Q) = \lambda_A(Q)V_A + \lambda_B(Q)V_B + \lambda_C(Q)V_C.$$

4.4 Interpretation

This is the biologically strongest ternary notebook in the folder for homologous protein sequences. Still, the plotted point should be read as a *relative allocation of normalized alignment similarity* to the three references, not as an exact Euclidean preservation of the three underlying dissimilarities.

5 Notebook 2: publication-quality MSA visualization of K-RAS homologs

5.1 Input design

The notebook `kras_msa_publication.ipynb` uses the FASTA file `kras_homologs_10.fa`, which contains ten homologous K-RAS proteins from UniProt. The first sequence is the human K-RAS protein and acts as the reference for alignment coordinates and residue numbering.

5.2 Reference-guided alignment construction

Let the reference sequence be $R = r_1, \dots, r_L$ and let $X_1, \dots, X_n$ be the homologous proteins. For each sequence X_i , the notebook computes a global pairwise alignment against R using BLOSUM62 and affine gap penalties. This yields:

- aligned residue assignments to each reference position,
- insertion strings between consecutive reference residues.

For each boundary $j \in \{0, \dots, L\}$, let $I_i(j)$ denote the insertion segment of sequence X_i between reference positions j and $j + 1$ (with $j = 0$ before the first reference residue and $j = L$ after the last one). The notebook defines the insertion capacity

$$m_j = \max_{i=1, \dots, n} |I_i(j)|.$$

The final multiple alignment is assembled by padding each insertion slot to length m_j . Thus each aligned sequence is written in a common coordinate system consisting of:

- an insertion block of width m_0 ,
- reference position 1,
- an insertion block of width m_1 ,
- reference position 2,
- and so on until reference position L and the tail insertion block of width m_L .

This is a reference-guided progressive construction rather than a full de novo MSA optimization, and it should be interpreted in the context of classical progressive-alignment ideas and later high-performance MSA heuristics [10, 9].

5.3 Consensus and conservation

If the aligned column k contains residue counts $n_k(a)$ over the non-gap alphabet, the notebook defines the consensus residue as the most frequent non-gap symbol and the conservation score as

$$c_k = \frac{\max_a n_k(a)}{\sum_a n_k(a)}.$$

Hence $c_k \in [0, 1]$, with $c_k = 1$ for a fully conserved non-gap column.

5.4 Visualization

The figure combines:

- residue-class coloring,
- monospace residue labels,
- a per-column conservation bar,
- a consensus row,
- and annotated K-RAS regions such as the P-loop, Switch I, Switch II, and the hypervariable C-terminal tail.

This notebook is intended for close inspection of a small homologous protein family, not for low-dimensional embedding.

6 Notebook 3: sequence similarity network from a FASTA file

6.1 Weighted similarity graph

The notebook `sequence_similarity_network.ipynb` starts from a FASTA file of homologous proteins and computes all pairwise normalized alignment similarities

$$w_{ij} = \frac{A(S_i, S_j)}{\sqrt{A(S_i, S_i) A(S_j, S_j)}},$$

where $A(S_i, S_j)$ is the global alignment score obtained from Biopython `PairwiseAligner` with BLOSUM62 and affine gaps.

This yields a symmetric similarity matrix

$$W = (w_{ij})_{i,j=1}^n, \quad w_{ii} = 1.$$

6.2 Thresholded sequence similarity network

For a chosen threshold $\tau \in [0, 1]$, the SSN is the weighted graph

$$G_\tau = (V, E_\tau), \quad V = \{1, \dots, n\},$$

with

$$(i, j) \in E_\tau \iff w_{ij} \geq \tau.$$

When an edge is present, its weight is the retained similarity itself:

$$\omega(i, j) = w_{ij}.$$

Thus the network is a filtered version of the full similarity matrix. Increasing τ produces a sparser graph and reveals finer neighborhoods; decreasing τ merges neighborhoods into larger connected components.

6.3 Distance matrix and cutoff selection

The notebook also forms the complementary distance matrix

$$D = (d_{ij})_{i,j=1}^n, \quad d_{ij} = 1 - w_{ij}.$$

Since the SSN cutoff is applied to the similarities w_{ij} , the corresponding distance cutoff is simply

$$d_{ij} \leq 1 - \tau.$$

Before plotting the network, the notebook ranks all unordered sequence pairs

$$\{(i, j) : 1 \leq i < j \leq n\}$$

by decreasing similarity, or equivalently by increasing distance. This ranked table is useful because it exposes the empirical distribution of pairwise relationships and makes it easier to identify a plausible threshold region.

In addition, the notebook visualizes the ordered distance matrix as a heatmap. If the sequences naturally separate into blocks or neighborhoods, these often appear as low-distance submatrices along the diagonal after suitable ordering. In practice, the heatmap, the pairwise ranking, and the empirical quantiles of $\{w_{ij}\}$ are used together to choose a sensible value of τ before drawing the final SSN.

6.4 Interpretation

This notebook is well suited to the exploration of local neighborhoods in sequence space, which is one of the main practical uses of SSNs in protein family analysis. Connected components, hubs, and dense subnetworks may indicate candidate subfamilies or groups of functionally related sequences.

However, the 2D graph layout used for plotting is only a visualization device. It is not a metric embedding and it is not a phylogenetic tree. This SSN use case follows the standard exploratory role of sequence similarity networks in protein-family analysis [3, 11, 17].

7 Notebook 4: ProteinMPNN on homologous PDB structures

7.1 Structure-conditioned design setup

The notebook `proteinmpnn_homologous_pdb.ipynb` starts from a seed protein sequence in FASTA format, queries the RCSB PDB for homologous structures, downloads a small representative set of entries, extracts one protein chain per structure, runs ProteinMPNN on each extracted backbone, and finally re-evaluates each generated sequence with repeated `score_only` passes. The workflow writes two FASTA outputs with distinct roles:

- `proteinmpnn_generated_sequences_design.fasta`: the raw design-stage ProteinMPNN output,
- `proteinmpnn_generated_sequences_reevaluated.fasta`: the same sequences after 100 repeated `score_only` evaluations.

The SSN notebook uses the second file by default.

If the homologous backbones are denoted by

$$B_1, \dots, B_m,$$

ProteinMPNN defines a conditional sequence model

$$p(\mathbf{s} \mid B_k),$$

where $\mathbf{s}$ is an amino-acid sequence compatible with the backbone geometry B_k .

7.2 Autoregressive inverse-folding model

In practice, ProteinMPNN is used as an inverse-folding model over protein backbones. For a decoding order π on the sequence positions, the conditional probability of a sequence can be written as

$$p(\mathbf{s} \mid B_k) = \prod_{t=1}^{L_k} p(s_{\pi(t)} \mid B_k, s_{\pi(1)}, \dots, s_{\pi(t-1)}),$$

where L_k is the chain length and B_k contains the geometric information of the target backbone. The model therefore converts backbone geometry into a residue-wise conditional distribution and samples new amino-acid identities from it [5].

7.3 Generated sequences and output scores

For each backbone B_k , the notebook samples one or more sequences

$$\mathbf{s}_{k,1}, \dots, \mathbf{s}_{k,r} \sim p(\mathbf{s} \mid B_k)$$

using the official ProteinMPNN implementation with default full-backbone weights. The output is therefore a collection of designed sequences conditioned on a family of homologous structures rather than on sequence similarity alone.

The raw design-stage output in `proteinmpnn_generated_sequences_design.fa` associates each sampled sequence with three summary quantities, following the official implementation [6]:

$$\text{score}_{\text{raw}}(\mathbf{s} \mid B_k) = -\frac{1}{|D_k|} \sum_{i \in D_k} \log p(s_i \mid B_k, \text{context}),$$

$$\text{global_score}_{\text{raw}}(\mathbf{s} \mid B_k) = -\frac{1}{L_k} \sum_{i=1}^{L_k} \log p(s_i \mid B_k, \text{context}),$$

and

$$\text{seq_recovery}(\mathbf{s}, \mathbf{s}^{\text{nat}}) = \frac{1}{L_k} \sum_{i=1}^{L_k} \mathbf{1}[s_i = s_i^{\text{nat}}].$$

Here D_k denotes the designed positions and $\mathbf{s}^{\text{nat}}$ is the native sequence associated with the backbone. In the current notebook workflow, each extracted target is reduced to a single designed chain, so the designed region effectively coincides with the whole chain and `score` and `global_score` are numerically identical in the exported summaries.

7.4 Repeated score-only re-evaluation

The one-pass quantities `scoreraw` and `global_scoreraw` reported during sampling are still stochastic because the ProteinMPNN forward pass injects random noise into the decoding context. To reduce that variance before the sequences are consumed by the SSN notebook, the workflow immediately re-scores every exported FASTA record with `ProteinMPNN --score_only` over

$$M = 100$$

independent stochastic passes. If $\hat{s}_{k,j}^{(m)}$ and $\hat{g}_{k,j}^{(m)}$ denote the m -th `score` and `global_score` evaluations of sequence $\mathbf{s}_{k,j}$ on backbone B_k , the notebook stores in `proteinmpnn_generated_sequences_reevaluat`

$$\bar{s}_{k,j} = \frac{1}{M} \sum_{m=1}^M \hat{s}_{k,j}^{(m)}, \quad \bar{g}_{k,j} = \frac{1}{M} \sum_{m=1}^M \hat{g}_{k,j}^{(m)},$$

and the corresponding empirical standard deviations

$$\sigma_s(k, j) = \sqrt{\frac{1}{M} \sum_{m=1}^M \left(\hat{s}_{k,j}^{(m)} - \bar{s}_{k,j} \right)^2}, \quad \sigma_g(k, j) = \sqrt{\frac{1}{M} \sum_{m=1}^M \left(\hat{g}_{k,j}^{(m)} - \bar{g}_{k,j} \right)^2}.$$

The re-evaluated FASTA therefore uses `score`= $\bar{s}_{k,j}$ and `global_score`= $\bar{g}_{k,j}$ as the downstream score fields, and also stores `score_std`, `global_score_std`, `score_runs`=100, `original_score`=`scoreraw`, and `original_global_score`=`global_scoreraw`. This gives the SSN notebook a more stable scalar attribute for node ranking and node-size encoding while preserving the original design-stage score for direct comparison.

7.5 Interpretation of good and bad scores

The ProteinMPNN `score` is an average negative log-probability per designed residue. Consequently, *lower is better*: a lower value means that the sampled amino acids are assigned higher probability by the model given the backbone. Conversely, larger values indicate that the sequence is less compatible with the backbone according to the learned model. In the present workflow, the primary quantity to interpret is the re-evaluated mean score $\bar{s}_{k,j}$ stored as `score` in `proteinmpnn_generated_sequences_reevaluated.fasta`; the raw one-pass score `scoreraw` is retained only as `original_score` for comparison.

There is no universal absolute cutoff separating “good” from “bad” designs across unrelated proteins, because the score depends on backbone class, chain length, model version, decoding temperature, and the design constraints used. The most defensible interpretation is therefore relative:

- compare designs produced for the same backbone or the same family of homologous backbones,
- prefer lower re-evaluated `score/global_score`,
- prefer smaller `score_std` when two designs have otherwise similar mean scores,
- prefer higher `seq_recovery` if the goal is native-like redesign rather than exploratory diversification.

The original ProteinMPNN paper reports a native-backbone sequence recovery of about 52.4% across their benchmark, which provides a useful reference scale for native-like redesign quality [5]. In the present notebook run on ten homologous K-RAS-related PDB backbones with two sampled designs per structure, the re-evaluated summary yields:

$$\overline{\text{design_score}} = 0.9087, \quad \overline{\text{global_score}} = 0.9087, \quad \overline{\text{design_score_std}} = 0.0150, \quad \overline{\text{seq_recovery}} = 0.459$$

The observed ranges are

$$\text{design_score} \in [0.8204, 0.9693], \quad \text{seq_recovery} \in [0.4152, 0.5361].$$

The increase from the original one-pass mean score to the re-evaluated mean score is expected: averaging 100 stochastic `score_only` passes removes favorable one-off fluctuations and yields a more conservative estimate of backbone compatibility. These values remain internally consistent with a reasonably strong redesign regime: the mean sequence recovery is close to the benchmark scale reported for native backbones, and the best samples combine lower mean scores, lower score dispersion, and higher native recovery. Within this workflow, designs with comparatively smaller $\bar{s}_{k,j}$, smaller $\sigma_s(k, j)$, and comparatively larger recovery should be regarded as the most promising candidates for downstream structural or functional validation.

7.6 Interpretation

This notebook is not a sequence-comparison method in the narrow sense. Instead, it asks how a homologous set of experimentally observed backbones constrains generative sequence design. It is therefore complementary to the SSN, MSA, and distance-based notebooks.

8 Notebook 5: ternary mapping from Hamming distance

8.1 Distance-to-similarity map

The notebook `ternary_hamming.ipynb` starts from

$$d_A(Q), \quad d_B(Q), \quad d_C(Q)$$

computed with normalized Hamming distance and transforms them into nonnegative similarity scores

$$s_A(Q) = \max(0, 1 - d_A(Q)), \quad s_B(Q) = \max(0, 1 - d_B(Q)), \quad s_C(Q) = \max(0, 1 - d_C(Q)).$$

8.2 Closure and barycentric mapping

The three scores are closed into

$$\lambda_i(Q) = \frac{s_i(Q)}{s_A(Q) + s_B(Q) + s_C(Q)}, \quad i \in \{A, B, C\},$$

and then mapped with the same barycentric triangle formula

$$p(Q) = \lambda_A(Q)V_A + \lambda_B(Q)V_B + \lambda_C(Q)V_C.$$

8.3 Interpretation

This notebook is a simple baseline. It is appropriate only when equal-length, position-wise comparison already captures the intended biological relation.

9 Notebook 6: ternary mapping from Levenshtein distance

The notebook `ternary_levenshtein.ipynb` has the same mathematical structure as the Hamming-based one, but replaces

$$d_i(Q) = d_{\text{Ham}}(Q, i)$$

with

$$d_i(Q) = d_{\text{Lev}}(Q, i), \quad i \in \{A, B, C\}.$$

Then it applies the same distance-to-similarity transform, the same closure, and the same barycentric map.

Its advantage over Hamming is robustness to unequal lengths and indels. Its disadvantage relative to alignment scoring is that all substitutions still have equal cost.

10 Why the ternary representation is limited

This part is central, because all three ternary notebooks are useful but mathematically limited.

10.1 1. They do not preserve the real sequence distances

The map

$$(d_A, d_B, d_C) \mapsto (s_A, s_B, s_C) \mapsto \left(\frac{s_A}{s_A + s_B + s_C}, \frac{s_B}{s_A + s_B + s_C}, \frac{s_C}{s_A + s_B + s_C} \right)$$

is many-to-one. Information is lost.

Therefore the triangle is not a metric embedding of sequence space.

10.2 2. The triangle encodes only relative proportions

Because

$$\lambda_A + \lambda_B + \lambda_C = 1,$$

the representation only preserves relative allocation among the three references.

Two sequences may have very different absolute distance scales and still end up at the same ternary position if their transformed similarity ratios are the same.

10.3 3. The result depends critically on the transform $d \mapsto s$

The geometry depends on how dissimilarities are converted to nonnegative similarities. Even if the Hamming and Levenshtein notebooks use

$$s = \max(0, 1 - d),$$

and the alignment notebook uses a normalized score

$$s = \frac{S(X, Y)}{\sqrt{S(X, X) S(Y, Y)}},$$

other transformations would generate different maps. Hence the representation is not invariant.

10.4 4. The map depends strongly on the chosen references

The three references define the coordinate system. Changing A , B , or C changes the whole ternary geometry. This is a landmark-type construction, not an intrinsic embedding of the full dataset, and in that sense it is conceptually closer to landmark-based ordination than to full-data embeddings [8].

10.5 5. The triangle is not a phylogenetic geometry

A point close to side AB does *not* imply that the sequence is evolutionarily between A and B . That visual pattern may arise from many causes: shared ancestry, convergence, saturation, or simply the chosen distance definition.

10.6 6. This is not strict CoDA on original data

In classical compositional data analysis, the parts are components of a common whole. Here the raw inputs are distances or similarities to three references, not literal parts of one total. The compositional structure is introduced by modeling choice after converting these quantities into nonnegative affinities and applying closure.

So the simplex viewpoint is appropriate for *relative similarity*, not because the original sequence measurements are themselves compositions.

11 Notebook 7: distance-based trilateration

11.1 Geometric idea

The notebook `trilateration_hamming.ipynb` takes the three reference points

$$V_A, \quad V_B, \quad V_C \in \mathbb{R}^2$$

and looks for a point $\mathbf{z}(Q)$ such that

$$\|\mathbf{z}(Q) - V_A\|_2 \approx d_A(Q), \quad \|\mathbf{z}(Q) - V_B\|_2 \approx d_B(Q), \quad \|\mathbf{z}(Q) - V_C\|_2 \approx d_C(Q).$$

This is the same mathematical structure as trilateration in localization problems.

11.2 Linearized equations

Starting from

$$\|\mathbf{z} - V_A\|_2^2 = d_A^2, \quad \|\mathbf{z} - V_B\|_2^2 = d_B^2, \quad \|\mathbf{z} - V_C\|_2^2 = d_C^2,$$

subtracting the first equation from the second and third yields

$$\begin{aligned} 2(V_B - V_A)^\top \mathbf{z} &= d_A^2 - d_B^2 + \|V_B\|_2^2 - \|V_A\|_2^2, \\ 2(V_C - V_A)^\top \mathbf{z} &= d_A^2 - d_C^2 + \|V_C\|_2^2 - \|V_A\|_2^2. \end{aligned}$$

This can be written as

$$M\mathbf{z} = \mathbf{r},$$

with

$$M = 2 \begin{bmatrix} (V_B - V_A)^\top \\ (V_C - V_A)^\top \end{bmatrix}.$$

The notebook solves this system by least squares.

11.3 Residuals

Since sequence distances are rarely perfectly realizable in a 2D Euclidean anchor system, the notebook also reports residuals

$$\rho_i(Q) = \left| \|\mathbf{z}(Q) - V_i\|_2 - d_i(Q) \right|, \quad i \in \{A, B, C\},$$

and their mean.

11.4 Interpretation

This approach is more faithful to actual distance magnitudes than ternary closure, but it still inherits the quality and limitations of the chosen distance function. In particular, if the distance model is too crude, the trilateration cannot repair that upstream weakness.

12 Notebook 8: classical MDS from the full distance matrix

12.1 Distance matrix

The notebook `mds_levenshtein.ipynb` uses all references and all queries together. If

$$\{S_1, \dots, S_N\} = \{A, B, C, Q_1, \dots, Q_n\},$$

then the pairwise distance matrix is

$$\Delta = (\delta_{ij})_{i,j=1}^N, \quad \delta_{ij} = d_{\text{Lev}}(S_i, S_j).$$

12.2 Double-centering step

Classical MDS builds a Gram-type matrix from squared distances [14]:

$$J = I_N - \frac{1}{N} \mathbf{1}\mathbf{1}^\top,$$

$$B = -\frac{1}{2} J \Delta^{\circ 2} J,$$

where $\Delta^{\circ 2}$ denotes elementwise squaring.

12.3 Spectral embedding

If

$$B = U \Lambda U^\top$$

is the eigendecomposition, then the 2D coordinates are

$$X_2 = U_2 \Lambda_2^{1/2},$$

where U_2 contains the first two eigenvectors associated with the largest nonnegative eigenvalues.

12.4 Interpretation

Unlike the ternary or trilateration notebooks, MDS is a global distance-based ordination. It does not privilege the three references except that they are included as part of the dataset [14].

Its target is not relative similarity to three anchors but approximate preservation of the full pairwise distance geometry. The same MDS construction could also be paired with a matrix of normalized alignment distances if protein-aware geometry is needed globally.

13 Notebook 9: PCA on clr-transformed k -mer compositions

13.1 From sequences to compositions

The notebook `kmer_pca.ipynb` does not start from edit distances. Instead, each sequence is converted into a composition of k -mer frequencies.

Let $\mathcal{A}$ be the alphabet and let

$$D = |\mathcal{A}|^k$$

be the number of possible k -mers. If $n_j(S)$ is the count of the j -th k -mer in sequence S , and $\alpha > 0$ is a pseudocount, define

$$\tilde{n}_j(S) = n_j(S) + \alpha.$$

The closed composition is

$$x_j(S) = \frac{\tilde{n}_j(S)}{\sum_{m=1}^D \tilde{n}_m(S)}.$$

Thus

$$\mathbf{x}(S) = (x_1(S), \dots, x_D(S))$$

lives in the simplex.

13.2 Centered log-ratio transform

To move from the simplex to a Euclidean coordinate system, the notebook applies the centered log-ratio transform

$$\text{clr}(\mathbf{x}) = \left(\log \frac{x_1}{g(\mathbf{x})}, \dots, \log \frac{x_D}{g(\mathbf{x})} \right),$$

where

$$g(\mathbf{x}) = \left(\prod_{j=1}^D x_j \right)^{1/D}$$

is the geometric mean.

13.3 PCA

Let

$$Z \in \mathbb{R}^{N \times D}$$

be the matrix whose rows are the clr vectors for all sequences. After column centering,

$$Z_c = Z - \mathbf{1}\bar{\mathbf{z}}^\top,$$

PCA is obtained from the singular value decomposition

$$Z_c = U\Sigma V^\top.$$

The first two principal component scores are

$$T_2 = Z_c V_2.$$

13.4 Interpretation

This notebook is not a distance embedding. It is a feature-based ordination that captures dominant variation in compositional sequence content. It is often preferable when alignments are unstable or when local compositional signal matters more than explicit sitewise homology.

14 Optional note on Aitchison geometry

The ternary coordinates

$$\boldsymbol{\lambda}(Q) = (\lambda_A(Q), \lambda_B(Q), \lambda_C(Q))$$

live in the simplex

$$\mathcal{S}^3 = \left\{ \mathbf{x} \in \mathbb{R}_+^3 : \sum_{i=1}^3 x_i = 1 \right\}.$$

If ternary points are compared quantitatively, the natural geometry is Aitchison geometry, not ordinary Euclidean geometry on raw parts [1, 18]. The Aitchison distance is

$$d_{\text{Ait}}(\mathbf{x}, \mathbf{y}) = \|\text{clr}(\mathbf{x}) - \text{clr}(\mathbf{y})\|_2.$$

This matters because Euclidean distances between plotted points in the triangle are not the same thing as compositional distances in the simplex.

15 When each notebook is appropriate

- Use `ternary_alignment_substitution.ipynb` as the default reference-based notebook for homologous protein sequences.
- Use `kras_msa_publication.ipynb` when the goal is to inspect residue-level conservation, local divergence, and functional regions in a small homologous protein family.
- Use `sequence_similarity_network.ipynb` when the main question is neighborhood structure, thresholded similarity connectivity, or subfamily detection in sequence space.
- Use `proteinmpnn_homologous_pdb.ipynb` when the goal is to generate backbone-conditioned sequences from a representative set of homologous PDB structures.
- Use `ternary_hamming.ipynb` only when all sequences have equal length and position-wise mismatches are already the intended comparison model.
- Use `ternary_levenshtein.ipynb` when indels and unequal lengths matter but you still want a simple reference-relative edit-distance baseline.
- Use `trilateration_hamming.ipynb` when you want a more literal anchor-based distance reconstruction in 2D.
- Use `mds_levenshtein.ipynb` when the global pairwise geometry of the whole dataset matters most.
- Use `kmer_pca.ipynb` when sequence composition and motif-like content are more relevant than direct alignment or edit distances.

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